

Heart Disease Risk Factor Analysis

Final Project Report | Python • SQL • Power BI

1. Objective

To perform an end-to-end analysis of patient health records in order to identify the demographic and clinical factors most strongly associated with heart disease, using a workflow spanning Python (data cleaning and exploratory analysis), SQL (business-question-driven querying), and Power BI (interactive dashboard reporting).

2. Dataset & Methodology

Dataset: UCI Heart Disease Dataset (Kaggle) | **Final Records:** 296 patients (after removing duplicates and invalid entries) | **Attributes:** 14

Data Cleaning: Removed duplicate records, verified column data types, and identified out-of-range placeholder values (ca = 4 and thal = 0) that did not match documented value ranges — these were removed as invalid entries rather than treated as valid data points.

Tools Used: Python (Pandas, Matplotlib, Seaborn) for cleaning and EDA; MS SQL Server for business-question analysis; Power BI for the interactive dashboard.

3. Key Findings from SQL Analysis

3.1 Overall Disease Prevalence

Status	Total Patients	Percentage
No Disease	136	45.95%
Disease	160	54.05%

3.2 Disease Rate by Age Group

Age Group	Total Patients	Disease Cases	Disease Rate
40-54	124	86	69.4%
55-64	117	43	36.8%
65+	41	21	51.2%
Under 40	14	10	71.4%

3.3 Disease Rate by Gender

Gender	Total Patients	Disease Cases	Disease Rate
Female	95	71	74.7%
Male	201	89	44.3%

3.4 Disease Rate by Chest Pain Type

Chest Pain Type	Total Patients	Disease Cases	Disease Rate
Type 0	141	39	27.7%
Type 1	49	40	81.6%
Type 2	83	65	78.3%
Type 3	23	16	69.6%

3.5 Cholesterol vs Disease Status

Status	Total Patients	Avg. Cholesterol
No Disease	136	251 mg/dl
Disease	160	243 mg/dl

3.6 Exercise-Induced Angina vs Disease

Angina on Exercise	Total Patients	Disease Cases	Disease Rate
No	199	137	68.8%
Yes	97	23	23.7%

3.7 Fasting Blood Sugar vs Disease

Fasting Blood Sugar	Total Patients	Disease Cases	Disease Rate
High	43	23	53.5%
Normal	253	137	54.2%

3.8 Resting Blood Pressure Category vs Disease

BP Category	Total Patients	Disease Cases	Disease Rate
Elevated (120-139)	139	79	56.8%
High (140+)	97	44	45.4%
Normal (Below 120)	60	37	61.7%

3.9 Maximum Heart Rate vs Disease Status

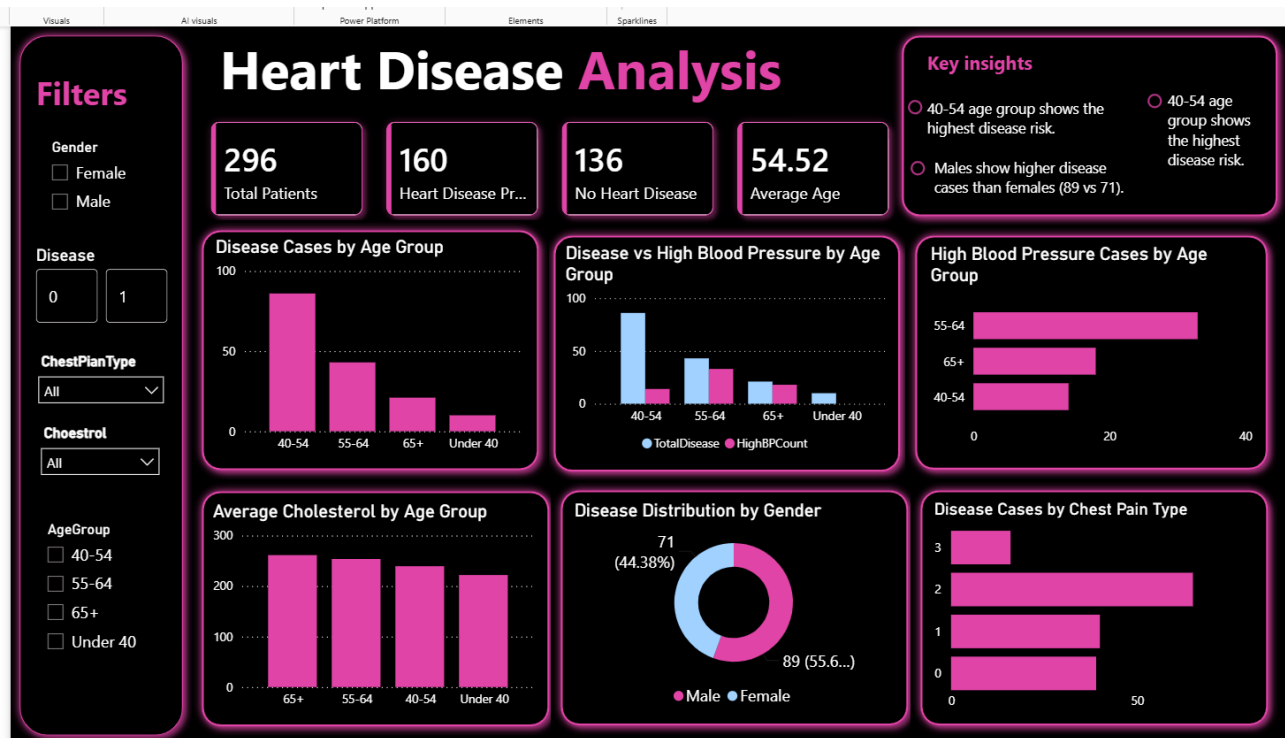
Status	Total Patients	Avg. Max Heart Rate
No Disease	136	138 bpm
Disease	160	158 bpm

3.10 Number of Major Vessels (ca) vs Disease

ca (Vessels)	Total Patients	Disease Cases	Disease Rate
0	173	129	74.6%
1	65	21	32.3%
2	38	7	18.4%
3	20	3	15.0%

4. Power BI Dashboard

An interactive dashboard was built to visualize the key metrics and patterns identified above, with slicers for Gender, Disease Status, Chest Pain Type, and Age Group for dynamic filtering.



Heart Disease Analysis — Interactive Power BI Dashboard

5. Key Insights

- Out of 296 patients analyzed, 160 (54.1%) have heart disease, indicating a fairly balanced but disease-leaning dataset.
- Chest pain types 1 and 2 show markedly higher disease rates (81.6% and 78.3% respectively) compared to type 0 (27.7%) — chest pain presentation is one of the strongest indicators in this dataset.
- The number of major vessels (ca) shows a clear inverse relationship with disease rate: patients with 0 vessels affected have a 74.6% disease rate, dropping steadily to 15.0% for patients with 3 vessels affected.
- Patients without exercise-induced angina show a notably higher disease rate (68.8%) than those with angina (23.7%) — a counter-intuitive but consistent pattern in this dataset that highlights the importance of validating clinical assumptions against actual data.
- Average cholesterol shows a counter-intuitive pattern: disease-positive patients actually have a slightly lower average cholesterol (243 mg/dl) than disease-negative patients (251 mg/dl), suggesting cholesterol alone is a weaker predictor in this dataset than commonly assumed.
- Fasting blood sugar shows almost no difference in disease rate between patients with high fasting sugar (53.5%) and normal fasting sugar (54.2%) — indicating it is not a meaningful differentiator for heart disease in this dataset.
- Resting blood pressure category also shows a counter-intuitive trend: patients with Normal blood pressure have the highest disease rate (61.7%), followed by Elevated (56.8%), while patients with High blood pressure (140+) show the lowest disease rate (45.4%) — indicating that in this dataset, disease risk does not increase directly with resting blood pressure category.

- Female patients in this dataset show a notably higher disease rate (74.7%) than male patients (44.3%), though males make up the larger share of total patients (201 vs 95).
- The 40-54 and Under-40 age groups show the highest disease rates (69.4% and 71.4%), while the 55-64 group shows a comparatively lower rate (36.8%) — indicating that risk in this dataset does not increase linearly with age.

6. Conclusion & Recommendations

This analysis shows that clinical presentation factors — chest pain type, number of major vessels affected, and exercise-induced angina — are more strongly associated with heart disease in this dataset than commonly assumed risk factors like cholesterol and resting blood pressure alone. Several patterns observed (such as the inverse relationship between exercise angina and disease, the declining disease rate with higher blood pressure category, and the non-linear relationship between age and risk) run counter to typical clinical expectations, underscoring the value of data-driven validation rather than relying on assumptions. For screening purposes, this suggests prioritizing chest pain characteristics and vessel imaging findings alongside traditional risk factors, and treating single-variable indicators like cholesterol or blood pressure alone with appropriate caution.